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CSES Problem Set

Two Knights

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Your task is to count for $k = 1, 2, \dots, n$ the number of ways two knights can be placed on a $k \times k$ chessboard so that they do not attack each other.

Input

The only input line contains an integer n .

Output

Print n integers: the results.

Constraints

- $1 \leq n \leq 10000$

Example

Input:

8

Output:

0

6

28

96

252

550

1056

1848

CSES Problem Set**Trailing Zeros**TASK [STATISTICS](#)**Time limit:** 1.00 s **Memory limit:** 512 MB

Your task is to calculate the number of trailing zeros in the factorial $n!$.

For example, $20! = 2432902008176640000$ and it has 4 trailing zeros.

Input

The only input line has an integer n .

Output

Print the number of trailing zeros in $n!$.

Constraints

- $1 \leq n \leq 10^9$

Example

Input:

20

Output:

4

CSES Problem Set

Gray Code

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

A Gray code is a list of all 2^n bit strings of length n , where any two successive strings differ in exactly one bit (i.e., their Hamming distance is one).

Your task is to create a Gray code for a given length n .

Input

The only input line has an integer n .

Output

Print 2^n lines that describe the Gray code. You can print any valid solution.

Constraints

- $1 \leq n \leq 16$

Example

Input:

2

Output:

00

01

11

10

CSES Problem Set**Tower of Hanoi**

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

The Tower of Hanoi game consists of three stacks (left, middle and right) and n round disks of different sizes. Initially, the left stack has all the disks, in increasing order of size from top to bottom.

The goal is to move all the disks to the right stack using the middle stack. On each move you can move the uppermost disk from a stack to another stack. In addition, it is not allowed to place a larger disk on a smaller disk.

Your task is to find a solution that minimizes the number of moves.

Input

The only input line has an integer n : the number of disks.

Output

First print an integer k : the minimum number of moves.

After this, print k lines that describe the moves. Each line has two integers a and b : you move a disk from stack a to stack b .

Constraints

- $1 \leq n \leq 16$

Example

Input:

2

Output:

3

1 2

1 3

2 3

CSES Problem Set**Creating Strings**

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Given a string, your task is to generate all different strings that can be created using its characters.

Input

The only input line has a string of length n . Each character is between a–z.

Output

First print an integer k : the number of strings. Then print k lines: the strings in alphabetical order.

Constraints

- $1 \leq n \leq 8$

Example

Input:

aabac

Output:

20

aaabc

aaacb

aabac

aabca

aacab

aacba

abaac

abaca

abcaa

acaab

acaba

acbba

baaac

baaca

bacaa

bcaaa

caaab

caaba

cabaa

cbaaa

CSES Problem Set

Apple Division

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

There are n apples with known weights. Your task is to divide the apples into two groups so that the difference between the weights of the groups is minimal.

Input

The first input line has an integer n : the number of apples.

The next line has n integers $p_1, p_2, \dots, p_n$: the weight of each apple.

Output

Print one integer: the minimum difference between the weights of the groups.

Constraints

- $1 \leq n \leq 20$
- $1 \leq p_i \leq 10^9$

Example

Input:

```
5
3 2 7 4 1
```

Output:

```
1
```

Explanation: Group 1 has weights 2, 3 and 4 (total weight 9), and group 2 has weights 1 and 7 (total weight 8).

CSES Problem Set**Chessboard and Queens**TASK [STATISTICS](#)**Time limit:** 1.00 s **Memory limit:** 512 MB

Your task is to place eight queens on a chessboard so that no two queens are attacking each other. As an additional challenge, each square is either free or reserved, and you can only place queens on the free squares. However, the reserved squares do not prevent queens from attacking each other.

How many possible ways are there to place the queens?

Input

The input has eight lines, and each of them has eight characters. Each square is either free (.) or reserved (*).

Output

Print one integer: the number of ways you can place the queens.

Example

Input:

```
.....  
.....  
..*.....  
.....  
.....  
.....**.  
...*.....  
.....
```

Output:

65

CSES Problem Set

Raab Game I

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Consider a two player game where each player has n cards numbered $1, 2, \dots, n$. On each turn both players place one of their cards on the table. The player who placed the higher card gets one point. If the cards are equal, neither player gets a point. The game continues until all cards have been played.

You are given the number of cards n and the players' scores at the end of the game, a and b . Your task is to give an example of how the game could have played out.

Input

The first line contains one integer t : the number of tests.

Then there are t lines, each with three integers n , a and b .

Output

For each test case print YES if there is a game with the given outcome and NO otherwise.

If the answer is YES, print an example of one possible game. Print two lines representing the order in which the players place their cards. You can give any valid example.

Constraints

- $1 \leq t \leq 1000$
- $1 \leq n \leq 100$
- $0 \leq a, b \leq n$

Example

Input:

```
5
4 1 2
2 0 1
3 0 0
2 1 1
4 4 1
```

Output:

```
YES
1 4 3 2
2 1 3 4
NO
YES
1 2 3
1 2 3
YES
1 2
2 1
NO
```

CSES Problem Set**Mex Grid Construction**

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Your task is to construct an $n \times n$ grid where each square has the smallest nonnegative integer that does not appear to the left on the same row or above on the same column.

Input

The only line has an integer n .

Output

Print the grid according to the example.

Constraints

- $1 \leq n \leq 100$

Example

Input:

5

Output:

```
0 1 2 3 4
1 0 3 2 5
2 3 0 1 6
3 2 1 0 7
4 5 6 7 0
```

CSES Problem Set

Knight Moves Grid

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

There is a knight on an $n \times n$ chessboard. For each square, print the minimum number of moves the knight needs to do to reach the top-left corner.

Input

The only line has an integer n .

Output

Print the number of moves for each square.

Constraints

- $4 \leq n \leq 1000$

Example

Input:

8

Output:

```
0 3 2 3 2 3 4 5
3 4 1 2 3 4 3 4
2 1 4 3 2 3 4 5
3 2 3 2 3 4 3 4
2 3 2 3 4 3 4 5
3 4 3 4 3 4 5 4
4 3 4 3 4 5 4 5
5 4 5 4 5 4 5 6
```

CSES Problem Set

Grid Coloring I

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

You are given an $n \times m$ grid where each cell contains one character A, B, C or D.

For each cell, you must change the character to A, B, C or D. The new character must be different from the old one.

Your task is to change the characters in every cell such that no two adjacent cells have the same character.

Input

The first line has two integers n and m : the number of rows and columns.

The next n lines each have m characters: the description of the grid.

Output

Print n lines each with m characters: the description of the final grid.

You may print any valid solution.

If no solution exists, just print IMPOSSIBLE.

Constraints

- $1 \leq n, m \leq 500$

Example

Input:

```
3 4
AAAA
BBBB
CCDD
```

Output:

```
CDCD
DCDC
ABAB
```

CSES Problem Set**Digit Queries**

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Consider an infinite string that consists of all positive integers in increasing order:

12345678910111213141516171819202122232425...

Your task is to process q queries of the form: what is the digit at position k in the string?

Input

The first input line has an integer q : the number of queries.

After this, there are q lines that describe the queries. Each line has an integer k : a 1-indexed position in the string.

Output

For each query, print the corresponding digit.

Constraints

- $1 \leq q \leq 1000$
- $1 \leq k \leq 10^{18}$

Example

Input:

3
3
7
19
12

Output:

7
4
1

CSES Problem Set

String Reorder

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

Your task is to reorder the characters of a string so that no two adjacent characters are the same. What is the lexicographically minimal such string?

Input

The only line has a string of length n consisting of characters A–Z.

Output

Print the lexicographically minimal reordered string where no two adjacent characters are the same. If it is not possible to create such a string, print -1 .

Constraints

- $1 \leq n \leq 10^6$

Example

Input:

HATTIVATTI

Output:

AHATITITVT

CSES Problem Set

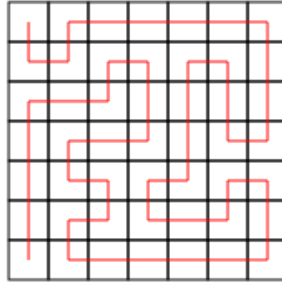
Grid Path Description

TASK STATISTICS

Time limit: 1.00 s **Memory limit:** 512 MB

There are 88418 paths in a 7×7 grid from the upper-left square to the lower-left square. Each path corresponds to a 48-character description consisting of characters D (down), U (up), L (left) and R (right).

For example, the path



corresponds to the description DRURRRRRDDDLUULDDDLDRRURDDLLLLLURULURRUULDLLDDDD.

You are given a description of a path which may also contain characters ? (any direction). Your task is to calculate the number of paths that match the description.

Input

The only input line has a 48-character string of characters ?, D, U, L and R.

Output

Print one integer: the total number of paths.

Example

Input:

?????R?????U????????????????????????????LD????D?

Output:

201